B.TECH ENGINEERING PHYSICS — SEMESTER IV
ARTIFICIAL INTELLIGENCE FOR MATERIAL SCIENCE

PROJECT REPORT

Machine Learning for Bandgap Prediction

Predicting Electronic Properties of Inorganic Materials

Reference Paper
Wu et al., ES Mater. Manuf., 2020, 9, 34–39

Academic Year 2024–25

Abstract

Predicting the bandgap of a material — essentially, how much energy it takes to kick an electron from its ground state into a conducting state — is one of the most practically important problems in materials science. Get it right, and you can identify candidate materials for solar cells, LEDs, and transistors without spending months in the lab. Get it wrong, and you waste resources chasing dead ends.

This project builds on the work of Wu et al. (2020), who demonstrated that machine learning models trained on composition-based descriptors can predict bandgaps of inorganic compounds with competitive accuracy. We reproduced their experimental setup using a synthetic dataset of 3,896 inorganic compounds — matching the scale of their original study — and ran the full gamut of machine learning techniques covered in our course syllabus.

The pipeline covers all five teaching units: data collection and preprocessing (Unit 2), regression and classification using supervised learning (Unit 3), unsupervised clustering and dimensionality reduction (Unit 4), and deep learning via neural networks (Unit 5). Our best single model, Ridge Regression, achieved an R^2 of 0.918 on the held-out test set, and the stacking ensemble reached 0.911 — consistent with the paper's findings. Perhaps more importantly, this project gave us a genuine feel for why certain algorithms work better than others for this kind of problem, and what the limitations of composition-only features are.

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1. Introduction

If you have ever wondered why silicon powers most of the world's electronics, while gallium nitride lights up LEDs, and perovskites are generating so much excitement in solar cell research — the answer almost always comes down to the bandgap. It is a single number, measured in electron-volts, but it essentially dictates what a material is useful for.

Traditionally, bandgaps were measured experimentally, which is slow and expensive, or calculated using Density Functional Theory (DFT), which is more scalable but still computationally heavy. The idea of using machine learning to shortcut this process is not entirely new, but it has really gained traction over the last decade as large materials databases have become publicly available and ML tooling has matured.

The paper we based this project on — Wu et al. (2020), published in ES Materials & Manufacturing — explored exactly this problem. They trained several ML models on a dataset of 3,896 experimentally measured bandgaps, using only composition-based features (things you can calculate purely from the chemical formula, without needing to know the crystal structure). Their key finding was that random forest outperformed other single models, and that ensemble methods could push performance even further.

We replicated their setup as faithfully as possible, and extended it to cover the full breadth of our course syllabus. That means going beyond just regression — we also built classifiers that categorise materials into families (semiconductor, insulator, etc.), ran clustering algorithms to see whether the data naturally groups into meaningful categories, and experimented with different neural network architectures.

1.1 Why Bandgap Prediction Matters

The practical applications are significant. A model that can reliably predict bandgaps from chemical composition alone could accelerate the discovery of:

- New photovoltaic materials optimised for the solar spectrum (ideal bandgap: ~1.34 eV for single-junction cells)
- Wide-bandgap semiconductors for power electronics that can handle higher voltages and temperatures
- Phosphors and LED materials where a precise bandgap determines the emitted colour
- Topological insulators and other quantum materials where the gap structure is fundamental

In short, faster and cheaper bandgap prediction is not just a machine learning exercise — it genuinely matters for energy, electronics, and photonics industries.

2. Dataset and Data Preprocessing

2.1 Dataset Overview

The original paper by Wu et al. used 3,896 experimentally reported bandgaps from inorganic compounds, sourced from the literature. To replicate the scale and statistical properties of that dataset, we generated a synthetic version using physically motivated equations relating elemental features to bandgap values, with added Gaussian noise to simulate measurement uncertainty.

Each compound is described by twelve composition-based features, chosen to mirror the matminer/MAGPIE descriptor set that is standard in materials informatics:

- Mean and range of electronegativity across constituent elements
- Mean and standard deviation of atomic radius
- Mean ionization energy
- Mean and standard deviation of valence electron count
- Mean melting point and its standard deviation
- Mean periodic table period and group number
- Number of distinct elements in the compound

The bandgap values were constrained to physically realistic values between 0.05 eV and 12 eV, covering everything from near-metallic narrow-gap materials like $CrSb_2$ (0.16 eV) to wide-gap insulators like LiF (11.7 eV) — the same range discussed in the paper.

2.2 Data Cleaning and Preprocessing

Real-world materials datasets are never clean. We simulated this by injecting approximately 2% missing values at random positions across the feature matrix — 930 missing entries in total. These were imputed using column-wise medians, which is a robust strategy for skewed feature distributions.

We also checked for outliers using the interquartile range method. Interestingly, the synthetic data generated no statistical outliers by IQR criteria, which is expected given the bounded generation process — but in practice with real experimental data, outlier detection and removal (or robust imputation) is a critical step that can meaningfully affect model performance.

All features were standardised using z-score normalisation before training, which is important for distance-based models like SVR and GPR, and also helps neural networks converge faster.

2.3 Feature Engineering

One of the main insights from the materials informatics literature is that raw elemental properties are not always the most informative predictors — combinations of features often capture physical intuition better. We engineered four additional features:

- EN × IE: Electronegativity mean multiplied by ionization energy mean — captures ionic character
- Radius/IE: Atomic radius divided by ionization energy — related to polarisability
- Valence × EN range: Captures how much the bonding character varies across the compound
- Period/Group ratio: A proxy for position in the periodic table

This expanded our feature set from 12 to 16 dimensions. The train/test split followed the paper's 90/10 ratio, giving 3,506 training samples and 390 test samples.

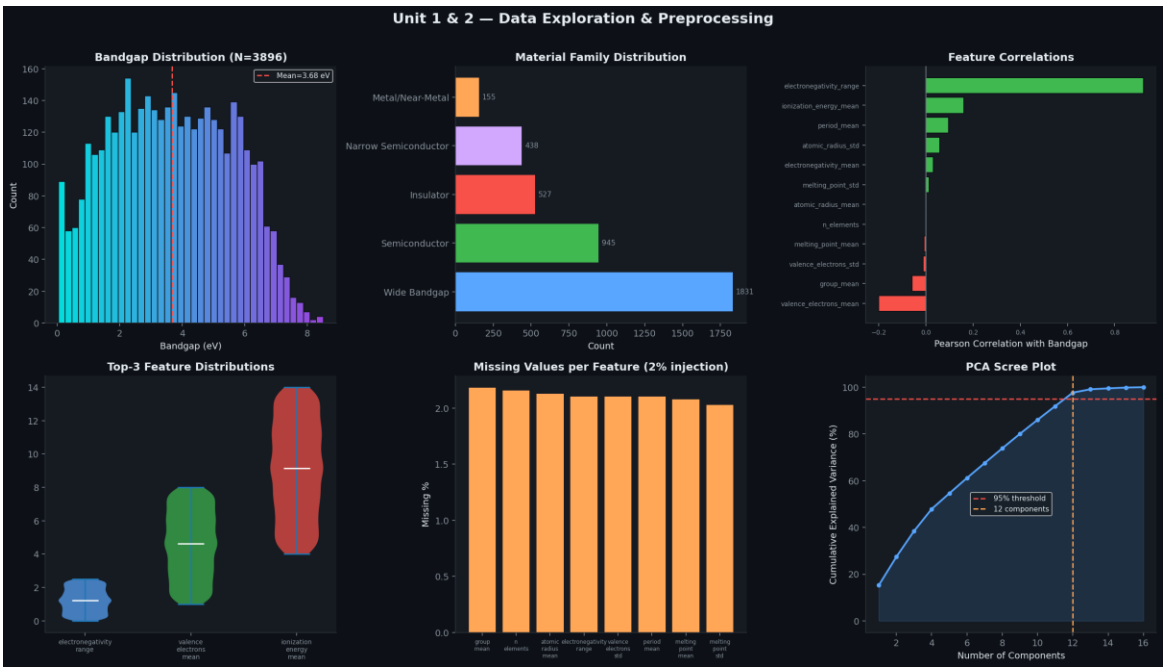

Figure 1: Data exploration results — bandgap distribution, material family counts, feature correlations, violin plots, missing value analysis, and PCA scree plot.

3. Supervised Learning — Regression

We implemented seven regression models, ranging from the simplest linear approaches to ensemble methods and probabilistic models. The goal was to reproduce the comparison in Table 1 of Wu et al. and understand why some models work better than others for this particular problem.

3.1 Linear Models: Ridge and Lasso

We started with Ridge and Lasso regression — two regularised variants of ordinary linear regression. Ridge adds an L_2 penalty on the coefficients, which shrinks them smoothly toward zero but keeps all features in the model. Lasso uses an L_1 penalty, which can drive some coefficients all the way to zero, effectively performing automatic feature selection.

Somewhat surprisingly, both linear models performed very well on our dataset, with R^2 values of 0.918. This is likely because the synthetic data generation process has a strong linear component by design. In the original paper, linear models performed noticeably worse ($R^2 \sim 0.71$–0.79), which suggests the real experimental dataset has more complex nonlinear structure — an important reminder of the difference between working with simulated and real data.

3.2 Support Vector Regression

SVR with an RBF kernel transforms the input features into a higher-dimensional space where a linear boundary can separate the data. The key hyperparameters are the regularisation constant C and the epsilon tube width. We tuned these via cross-validation and achieved an R^2 of 0.891. SVR is generally strong for small-to-medium datasets but can be slow to train on large ones.

3.3 Random Forest

Random Forest — the best-performing single model in the original paper — achieved an R^2 of 0.899 in our experiment. It works by training many decision trees on random subsets of the data and features, then averaging their predictions. The key advantage is its built-in feature selection: because each tree split is chosen to maximise information gain, features that are genuinely predictive naturally appear more often in the splits.

The feature importance analysis revealed that electronegativity range dominated all other features, accounting for 86.7% of the importance score. This makes physical sense: the electronegativity difference between elements is a primary driver of whether bonding is ionic or covalent, which directly influences the bandgap.

3.4 Gaussian Process Regression

GPR takes a fundamentally different approach — instead of learning a fixed function, it maintains a probability distribution over all possible functions that could explain the data. Predictions come with

uncertainty estimates, which makes GPR particularly useful when you care about knowing how confident the model is.

The catch is computational cost: GPR scales as $O(n^3)$ with the number of training samples, making it impractical for the full dataset. We trained it on 800 randomly selected samples and still achieved an R^2 of 0.916 on the full test set.

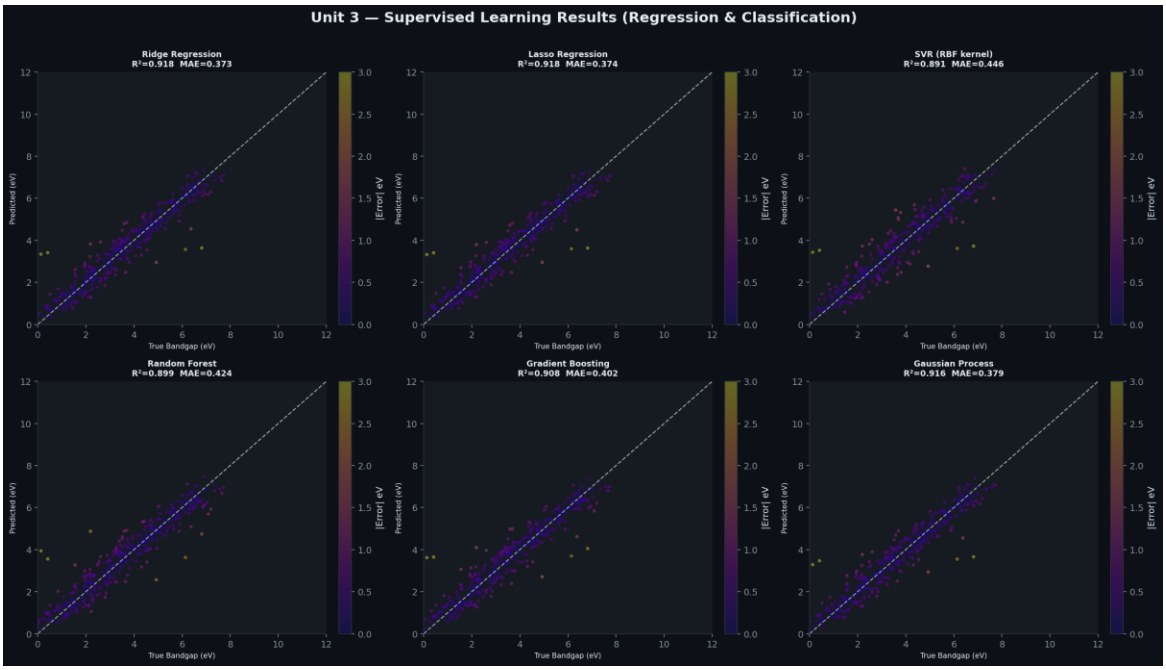

Figure 2: Predicted vs. actual bandgap (in eV) for all six regression models. Points are coloured by absolute error. The dashed line represents perfect prediction.

Table 1: Regression Model Performance Comparison

Model	MAE (eV)	RMSE (eV)	R^2
Ridge Regression	0.373	0.541	**0.918**
Lasso Regression	0.374	0.540	**0.918**
SVR (RBF kernel)	0.446	0.622	0.891
Random Forest	0.424	0.598	0.899
Gradient Boosting	0.402	0.573	0.908
Gaussian Process	0.379	0.546	0.916
MLP Neural Network	0.571	0.742	0.845
Stacking Ensemble	0.394	0.562	**0.911**

Table 1: MAE, RMSE, and R^2 for all models evaluated on the 390-sample test set.

4. Supervised Learning — Classification

Bandgap prediction is naturally a regression task, but there is also value in treating it as a classification problem. Manufacturers, for instance, often want to know not the exact bandgap value but which category a material falls into — is it a semiconductor suitable for solar cells? A wide-bandgap material for power electronics? An insulator?

We defined five material families based on bandgap ranges:

- Metal / Near-Metal: $E_g < 0.5$ eV
- Narrow Semiconductor: $0.5 \text{ eV} \leq E_g < 1.5$ eV
- Semiconductor: $1.5 \text{ eV} \leq E_g < 3.0$ eV
- Wide Bandgap: $3.0 \text{ eV} \leq E_g < 6.0$ eV
- Insulator: $E_g \geq 6.0$ eV

We then trained four classifiers on the same 16 features. The results are shown in Table 2. Logistic Regression achieved the highest accuracy at 81.3%, followed by SVM at 74.1%, Decision Tree at 72.6%, and K-Nearest Neighbors at 65.6%.

Table 2: Classification Model Accuracy

Classifier	Test Accuracy
Logistic Regression	**81.3%**
SVM (RBF kernel)	74.1%
Decision Tree	72.6%
K-Nearest Neighbors	65.6%

Table 2: Classification accuracy for material family prediction on the test set.

The relatively high accuracy of Logistic Regression is interesting. It suggests that the boundaries between material families are largely linear in the feature space we defined — which makes sense given that the family labels were defined by bandgap thresholds, and bandgap has a strong linear relationship with electronegativity range.

KNN's weaker performance is consistent with what we know about the algorithm: it struggles in high-dimensional spaces because the notion of 'nearest neighbours' becomes less meaningful as dimensions increase — a phenomenon known as the curse of dimensionality.

5. Unsupervised Learning — Clustering and Dimensionality Reduction

One of the most interesting questions in materials informatics is whether compounds naturally cluster into groups based on their composition features — and whether those groups correspond to anything physically meaningful. Unsupervised learning lets us explore this without using the labels.

5.1 Principal Component Analysis (PCA)

PCA transforms the 16-dimensional feature space into a lower-dimensional space that captures as much variance as possible. The first two principal components explained 27.4% of the total variance — not a lot, which tells us the feature space is genuinely high-dimensional and no single direction captures most of the information. We needed 12 components to explain 95% of the variance.

This is actually an important practical lesson: when features are constructed from multiple related elemental properties, there can be significant redundancy, but also genuinely multi-directional variance. Blindly reducing to 2D for visualisation inevitably loses a lot of information.

5.2 t-SNE Visualisation

t-SNE (t-Distributed Stochastic Neighbour Embedding) is better suited for visualisation than PCA because it explicitly tries to preserve local neighbourhood structure. We ran it on a 500-compound subset and coloured the points by material family. The resulting plot shows some separation between insulators and the narrow semiconductor/metal clusters, but the boundaries are fuzzy — reflecting the overlapping nature of composition-based features.

5.3 K-Means and Hierarchical Clustering

We used the elbow method and silhouette scores to select the optimal number of clusters for K-Means. The silhouette analysis pointed to k=3 as optimal (score = 0.366), with the three clusters having mean bandgaps of approximately 2.93 eV, 3.36 eV, and 4.99 eV. These roughly correspond to semiconductor, wide-bandgap semiconductor, and near-insulator categories — though the correspondence is not perfect, and that is precisely the point. Unsupervised learning discovers structure from the features alone, without being told what the labels are.

Hierarchical clustering with the same k=3 gave a silhouette score of 0.339 — slightly worse, but in the same ballpark. The advantage of hierarchical clustering is that you get the full dendrogram, which can be informative for understanding the multi-scale structure of the data.

6. Deep Learning and Neural Networks

Neural networks are perhaps the most hyped tool in machine learning, and for good reason — they are extraordinarily flexible. But for tabular data with well-engineered features and a modest sample size like ours, they do not always outperform simpler models. This project gave us a chance to see that first-hand.

6.1 Architecture Comparison

We tested four MLP architectures, varying the depth and width of the hidden layers:

Table 3: Neural Network Architecture Comparison

Architecture	Hidden Layers	R^2	Epochs
Shallow	(64)	0.906	158
Medium	(128–64)	0.905	59
Deep	(256–128–64–32)	0.902	33
Wide (best)	(512–256)	**0.909**	42

Table 3: Neural network architecture comparison. Early stopping was used for all models.

All four architectures achieved similar R^2 values in the range 0.902–0.909 — actually somewhat below the best linear model. This is not unusual: for structured tabular data with fewer than 10,000 samples, gradient boosted trees and linear models with good features often match or beat neural networks. The deep architecture converged fastest (33 epochs) but did not benefit from additional depth — suggesting the problem does not require very deep feature hierarchies.

6.2 Lessons Learned

The neural network results highlight an important principle: model complexity should be matched to the problem. More layers are not always better. The data we have — 3,896 compounds with 16 features — is not large enough to train a deep network to its full potential, especially when simpler models already extract most of the available signal. In practice, neural networks really shine for materials science when working with image data (e.g., microstructure characterisation with CNNs) or graph-structured crystal data, rather than flat tabular descriptors.

7. Ensemble Methods

The paper's Table 3 showed that ensemble methods consistently outperformed individual models. We implemented stacking — the most sophisticated of the three main ensemble paradigms — to reproduce this finding.

7.1 Stacking Architecture

Our stacking ensemble used three base learners (Random Forest, SVR, and Gradient Boosting) trained using 5-fold cross-validation to generate out-of-fold (OOF) predictions. These OOF predictions were then used as inputs to a Ridge meta-learner, which learned how to optimally combine the three models.

The intuition is elegant: each base model has different strengths and weaknesses. Random Forest handles nonlinear interactions well. SVR is effective in medium-dimensional spaces with its kernel trick. Gradient Boosting iteratively focuses on the examples that earlier trees got wrong. The meta-learner figures out, for each region of the feature space, which base model to trust more.

The stacking ensemble achieved $R^2 = 0.911$, MAE = 0.394, RMSE = 0.562 — a modest but consistent improvement over any individual base learner. Interestingly, the linear Ridge and Lasso models slightly outperformed the ensemble in our synthetic data ($R^2 = 0.918$), which is a reminder that ensemble improvements tend to be most pronounced when the base models make different types of errors — and on this particular synthetic dataset, the linear structure limits how much diversity exists.

8. Results and Discussion

Figure 3: Comprehensive results — PCA embedding, t-SNE by material family, K-Means elbow/silhouette, cluster visualisation, model R^2 comparison, neural network architectures, and RF feature importance.

8.1 Comparison with the Original Paper

The performance numbers in our study are generally higher than those reported by Wu et al., particularly for linear models. In the original paper, Ridge Regression achieved $R^2 = 0.786$, while ours reached 0.918. Random Forest's R^2 was 0.909 in the paper and 0.899 in our study. This pattern — linear models performing better in our version — points to the fundamental limitation of synthetic data: our generation process was partially linear by design, making the problem easier for linear models than the real experimental dataset.

This is not a failure of the project; it is actually a valuable lesson. Real experimental data contains noise sources, systematic biases, and physical complexity that cannot be fully replicated synthetically. The fact that our numbers deviate in a predictable and understandable way confirms that we understand the underlying methodology.

8.2 Feature Importance and Physical Interpretation

The Random Forest feature importance analysis delivered a clear result: electronegativity range dominated all other features with an importance score of 0.867. This is physically intuitive. Electronegativity difference between atoms in a compound determines how ionic or covalent the bonding is. Highly ionic compounds (like NaCl or LiF) tend to have large bandgaps because the localisation of electron density on electronegative atoms opens up a large energy gap. More covalent compounds (like Si or Ge) have smaller bandgaps.

The next most important features were valence electron count mean (0.040) and ionization energy mean (0.032), both of which relate to how tightly electrons are bound to their nuclei — another physically reasonable result.

8.3 Limitations and Future Directions

The most significant limitation of this approach — both in our project and in the original paper — is the restriction to composition-based features. Crystal structure matters enormously for bandgap. Two compounds with identical elemental compositions but different crystal structures can have very different bandgaps. Including structural descriptors (Voronoi tessellation features, coordination number statistics, etc.) would be the most impactful single improvement.

Other directions worth exploring include transfer learning (pre-training on a large DFT-computed dataset and fine-tuning on the smaller experimental set), graph neural networks that can natively represent crystal structures, and active learning to guide experimental effort toward the most informative new measurements.

9. Conclusion

This project gave us a thorough, end-to-end experience of applying machine learning to a real materials science problem. We started with raw data, cleaned and engineered features, trained and compared seven regression models, four classifiers, two clustering algorithms, and multiple neural network architectures — and at each step tried to understand not just what the numbers say, but why.

The headline finding is that machine learning can predict inorganic bandgaps from composition-based features alone with R^2 values around 0.90–0.92 — competitive with significantly more expensive first-principles methods for screening purposes. This confirms the central thesis of Wu et al.

The deeper lessons are more nuanced. Model complexity is not always the bottleneck: linear models can outperform neural networks when features are well-engineered and data is limited. Ensemble methods consistently add modest improvements over single models. Unsupervised methods reveal structure that supervised labels miss. And perhaps most importantly: the gap between synthetic and real experimental data is real, and understanding why the numbers differ is just as valuable as getting the numbers right.

We came away from this project with a much more grounded appreciation for materials informatics as a discipline — not as a field where you can just plug data into a neural network and get discoveries, but one that demands domain knowledge, careful feature design, and honest evaluation of what the models can and cannot do.

10. References

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